

Research Methods

Day 12 · Session 12: Research Ethics, Final Slide Polish & References

Outline for Today

1. **Research Ethics — Why It Matters**
2. **'Cite That Right' Team Activity**
3. **Slide Design Workshop**
4. **Full Run-Through**



Session 12: Research Ethics, Slide Polish & References

Learning Objectives

Understand core principles of research ethics

Attribution, honesty, limitations, and consent — the values that make research trustworthy.

Complete the References slide with proper citations

Every source cited, every citation consistent, AI use disclosed.

Polish slides for visual clarity and professional appearance

One idea per slide, clean design, findings-first chart titles.

Practice a full presentation run-through for the first time

End to end, timed, with focused peer and instructor feedback.



Morning: ResearchCORE



Research Ethics – Why It Matters

Why Ethics Matters

Research earns trust only when it is done honestly. One dishonest study can undo the credibility of an entire field.

As a researcher you make promises — to your participants, your readers, and the people your findings might affect.

Four pillars we'll return to today:

- **Attribution** — credit every source and tool
- **Honesty** — report what you found, not what you hoped
- **Limitations** — own what your study can't say
- **Consent** — respect the people behind the data

The stakes are real

Findings become policy, medicine, and code. Someone will act on what you publish — ethics is how you make sure that action is well-founded.

Core Ethical Considerations

Respect Privacy

Protect participants' identities and personal information. Use anonymous responses whenever possible.

Informed Consent

Ensure participants understand what they are agreeing to. Get permission from parents for minors.

Minimize Harm

Design research that does not cause physical or emotional distress to participants.

Get Approvals

Obtain necessary permissions from teachers, administrators, or other authorities.

Plagiarism & Proper Attribution

Plagiarism = using someone's words, ideas, data, or figures without credit — whether on purpose or by accident.

Give credit for:

- Direct quotes *and* paraphrased ideas
- Data, tables, and figures you didn't create
- Methods or frameworks you borrowed

✓ The safe habit

When in doubt, cite. A citation never weakens your work — it shows you know the landscape and stand on solid ground.

“ Quote vs. paraphrase

Quoting copies exact words (in quotation marks); paraphrasing restates in your own words. **Both** still need a citation.

AI Ethics – What Do You Have to Disclose?

Using AI tools is allowed – **hiding it is not**. Major journals (Nature, Science, APA) now require disclosure.

Disclose:

- Which AI tools you used
- What you used them *for* (coding, editing, brainstorming)
- That you **verified** the outputs yourself

A sample attribution statement

"AI tools were used in this project for code drafting and language editing. All analyses were conducted and verified by the research team. All interpretations and conclusions are our own."

 **You** remain responsible for every number and claim – the AI is a tool, not an author.

Researcher Bias – and How to Minimize It

Bias is the pull toward the answer you *want* to find. Everyone has it; good researchers guard against it.

Common forms:

- **Confirmation bias** — noticing only evidence that fits
- **Selection bias** — a sample that isn't representative
- **Funding/interest bias** — a stake in the outcome

Guardrails

- Pre-decide your analysis *before* seeing results
- Report findings that **don't** fit your hypothesis
- Show your full data and code
- Invite someone to challenge your interpretation

Overstating Findings: p-Hacking & Cherry-Picking

p-hacking

Trying test after test until something crosses $p < 0.05$, then reporting only that one. It manufactures "significance" that won't replicate.

Cherry-picking

Reporting only the results, cases, or time-windows that support your story — and quietly dropping the rest.

✓ **The honest alternative:** report all the analyses you ran, use hedged language (*suggests, is associated with*), and let a non-significant result stand as a finding. A true null is not a failure.

The Responsibility of Publishing

Once your findings are public, **you no longer control how they're used.**

Ask before you publish:

- Who might **act** on this result?
- Could it be **misread** or **misused** to harm a group?
- Have I stated my **limitations** loudly enough to prevent overreach?

Frame findings carefully

A single correlation, stripped of caveats, can become a headline or a policy. Precise, honest framing is part of your ethical duty — not an afterthought.

When Research Goes Wrong



Tuskegee Study

For 40 years, treatment was withheld from Black men with syphilis without their informed consent — the case that reshaped U.S. consent rules.



Data Fabrication

Fabricated results in psychology helped fuel the **replication crisis** — famous findings that collapsed when others tried to reproduce them.



AI Facial Recognition Bias

Systems trained on skewed data misidentified women and darker-skinned faces at far higher rates — biased data, biased harm.

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The common thread: each failure came from ignoring one of our pillars — consent, honesty, or attribution of the data's limits.

”

'Cite That Right' Team Activity

'Cite That Right' – Team Activity (20 min)

Complete the '**Cite That Right**' Handout for your full reference list. Run these four checks:

1

Every source **cited in your slides** appears in References.

2

Every source **in References** was actually cited in a slide.

3

Consistent citation style throughout (pick one and hold it).

4

AI tools documented with an attribution statement.

☰ In-text → reference list

In a slide: "...failing to plan is planning to fail (Yin & Ogut, 2025)."

In References: Yin & Ogut (2025). *Nexellence ResearchCORE: An Introduction to Research Methods*. Nexellence Press.

Toska, G. (2023). *Counseling Centers in Türkiye*. National Conference on Counseling Psychology, Istanbul.

20:00



Slide Design Workshop

Slide Design Principles



One idea per slide. If a slide makes two points, split it.

6×6

The 6×6 rule: at most 6 bullets, 6 words each. Slides are cues, not scripts.



Chart titles are findings, not labels:
"Wages rise with experience" > "Wage vs. experience".

A

Two fonts, maximum. One for headings, one for body.



High-contrast colors. Dark text on light, or light on dark — never mid-gray on white.



Accessible palettes. Color-blind-safe; never rely on color alone to carry meaning.

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 **Read every slide from the back of the room.** If you can't, your audience can't either.

The Slide Design Checklist

Trade slides with another team and score each one:

- One clear idea, stated in the title
- ≤ 6 bullets, ≤ 6 words each
- Chart title states the **finding**
- Axis labels legible at slide size
- ≤ 2 fonts; consistent throughout
- High-contrast, color-blind-safe
- Sources cited on the slide

Peer review, out loud

For each slide your partner team can't instantly "get," ask: *what is the one thing this slide is trying to say?* If it takes more than a sentence to answer, the slide needs work.

 When in doubt, **cut**. Less on the slide means more attention on you.

Slide Showdown

Put your work up for a friendly, fast critique:

 **Show 2 slides of your *best* work**

What makes them land? Name the design choice so the class can steal it.

 **Show 2 slides you're *unsure* about**

The class suggests one concrete fix each.
No slide leaves without an action.

30:00

Reference: The Presentation Outline

Slide	What it contains
Title	Project title, presenters, date, institution, logos
Topic	The global challenge; define it and say why it matters (cite sources)
Literature Review	What's known, arranged by theme (cite sources)
Research Question	Your question(s) and, if possible, a hypothesis
Methods	Secondary analysis; data source & year; sample size & traits; key variables; tools
Results	Descriptive stats per variable, then figures/tables (may be several slides)
Discussion	How you interpret what you found
Conclusion	Final takeaway and close
References	Full list, one consistent style

 Every polished deck follows this arc. If a slide doesn't fit the arc, ask whether it belongs.



Full Run-Through

Full Run-Through (40 min)

Each team presents the **entire** presentation, beginning to end — the first real rehearsal.



Time it: target **10–12 minutes** per team.



Class observes silently — take notes, hold questions.



3 notes per team from the instructor.

+ 2 strengths

What worked — keep doing it.

🔧 1 specific improvement

One concrete, actionable change for next time — not a vague "be clearer."

12:00

What to Listen For



Timing

Does it fit 10–12 minutes?
Where does it drag or rush?



Transitions

Do the hand-offs between
speakers and sections feel
smooth?



Takeaway

Could you state the team's main
finding in one sentence
afterward?

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🌱 A first run-through is *supposed* to be rough. Its job is to surface the fixes while there's still time to make them.

Session 12: What You Can Do Now

Research ethics:

- Explain the four pillars — **attribution, honesty, limitations, consent** — and spot p-hacking, cherry-picking, and researcher bias in your own work.
- Name why historical cases (Tuskegee, fabrication in the replication crisis, AI facial-recognition bias) still shape research rules today.

References & attribution:

- Complete a **References slide** where every citation matches a source in the text, in one consistent style, with AI use disclosed.

Slide polish & delivery:

- Apply the design principles — one idea per slide, the 6×6 rule, findings-first chart titles, ≤ 2 fonts, high-contrast accessible color — and review a peer's deck with the checklist.
- Deliver a full, timed run-through and turn feedback (2 strengths + 1 fix) into concrete revisions.